

CSE5095 Homework 2: Type Safety

Your Name

Booleans

Task 1

Values.

$$\frac{}{\text{true val}} \text{VALTRUE} \qquad \frac{}{\text{false val}} \text{VALFALSE}$$

Steps.

$$\frac{e_1 \mapsto e'_1}{\text{if } e_1 \text{ then } e_2 \text{ else } e_3 \mapsto \text{if } e'_1 \text{ then } e_2 \text{ else } e_3} \text{STEPIFSEARCH}$$
$$\frac{}{\text{if true then } e_2 \text{ else } e_3 \mapsto e_2} \text{STEPIFTRUE} \qquad \frac{}{\text{if false then } e_2 \text{ else } e_3 \mapsto e_3} \text{STEPIFFALSE}$$

Only the condition is searched; the branches are never evaluated until one is selected.

Task 2

$$\begin{array}{ll} \text{if } 1 + 2 = 4 \text{ then "Fake news" else "True fact"} & \\ \mapsto \text{if } 3 = 4 \text{ then "Fake news" else "True fact"} & (\text{STEPIFSEARCH}) \\ \mapsto \text{if false then "Fake news" else "True fact"} & (\text{STEPIFSEARCH}) \\ \mapsto \text{"True fact"} & (\text{STEPIFFALSE}) \end{array}$$

The derivations of the first two steps are:

$$\frac{\frac{1 + 2 \mapsto 3}{1 + 2 = 4 \mapsto 3 = 4} \text{STEPADD}}{\text{if } 1 + 2 = 4 \text{ then "Fake news" else "True fact"} \mapsto \text{if } 3 = 4 \text{ then "Fake news" else "True fact"}} \text{STEPSEARCHEQLEFT}$$
$$\frac{\frac{3 \neq 4}{3 = 4 \mapsto \text{false}} \text{STEPNOTEQ}}{\text{if } 3 = 4 \text{ then "Fake news" else "True fact"} \mapsto \text{if false then "Fake news" else "True fact"}} \text{STEPIFSEARCH}$$

and "True fact" val, so evaluation is complete.

Task 3

Abbreviations: TI = TYPEIF, TE = TYPEEQ, TP = TYPEPLUS, TN = TYPENUM, TS = TYPESTR.

$$\frac{\frac{\frac{}{1 : \text{int}} \text{TN} \quad \frac{}{2 : \text{int}} \text{TN}}{1 + 2 : \text{int}} \text{TP} \quad \frac{}{4 : \text{int}} \text{TN} \quad \frac{}{\text{"Fake news" : string}} \text{TS} \quad \frac{}{\text{"True fact" : string}} \text{TS}}{1 + 2 = 4 : \text{bool} \quad \text{if } 1 + 2 = 4 \text{ then "Fake news" else "True fact" : string} \text{TI}} \text{TE}$$

Task 4

Preservation. If $e : \tau$ and $e \mapsto e'$, then $e' : \tau$.

Proof. By induction on the derivation of $e \mapsto e'$. New cases:

- **Case StepIfTrue.** Then $e = \text{if true then } e_2 \text{ else } e_3$ and $e' = e_2$. The only typing rule whose conclusion has the form $\text{if } \cdot \text{ then } \cdot \text{ else } \cdot$ is TYPEIF, so by inversion $\text{true} : \text{bool}$, $e_2 : \tau$, and $e_3 : \tau$. In particular $e' = e_2 : \tau$.
- **Case StepIfFalse.** Then $e = \text{if false then } e_2 \text{ else } e_3$ and $e' = e_3$. By inversion on TYPEIF, $e_3 : \tau$, i.e. $e' : \tau$.
- **Case StepIfSearch.** Then $e = \text{if } e_1 \text{ then } e_2 \text{ else } e_3$, $e' = \text{if } e'_1 \text{ then } e_2 \text{ else } e_3$, and $e_1 \mapsto e'_1$. By inversion on TYPEIF, $e_1 : \text{bool}$, $e_2 : \tau$, and $e_3 : \tau$. Since $e_1 : \text{bool}$ and $e_1 \mapsto e'_1$, the induction hypothesis gives $e'_1 : \text{bool}$. Then by TYPEIF, $\text{if } e'_1 \text{ then } e_2 \text{ else } e_3 : \tau$, i.e. $e' : \tau$. $\square$

Task 5

Canonical Forms (Booleans). If $e : \text{bool}$ and $e \text{ val}$, then $e = \text{true}$ or $e = \text{false}$.

Task 6

Progress. If $e : \tau$, then $e \text{ val}$ or there exists e' such that $e \mapsto e'$.

Proof. By induction on the derivation of $e : \tau$. New cases:

- **Case TypeTrue.** Then $e = \text{true}$, and true val by VALTRUE.
- **Case TypeFalse.** Then $e = \text{false}$, and false val by VALFALSE.
- **Case TypeIf.** Then $e = \text{if } e_1 \text{ then } e_2 \text{ else } e_3$ with $e_1 : \text{bool}$, $e_2 : \tau$, $e_3 : \tau$. By the induction hypothesis on $e_1 : \text{bool}$, either $e_1 \text{ val}$ or $e_1 \mapsto e'_1$ for some e'_1 .
 - If $e_1 \mapsto e'_1$, then by STEPIFSEARCH, $e \mapsto \text{if } e'_1 \text{ then } e_2 \text{ else } e_3$.
 - If $e_1 \text{ val}$, then since $e_1 : \text{bool}$, Canonical Forms (Task 5) gives $e_1 = \text{true}$ or $e_1 = \text{false}$. If $e_1 = \text{true}$, then $e \mapsto e_2$ by STEPIFTRUE. If $e_1 = \text{false}$, then $e \mapsto e_3$ by STEPIFFALSE.

In every subcase e steps. $\square$

Division by zero

Task 7

$$\begin{array}{c}
\frac{}{n : \mathbf{mz}} \text{TYPENUM} \qquad \frac{n \neq 0}{n : \mathbf{nz}} \text{TYPENUMNZ} \\
\\
\frac{e_1 : \mathbf{mz} \quad e_2 : \mathbf{mz}}{e_1 + e_2 : \mathbf{mz}} \text{TYPEADD} \qquad \frac{e_1 : \mathbf{mz} \quad e_2 : \mathbf{mz}}{e_1 - e_2 : \mathbf{mz}} \text{TYPESUB} \\
\\
\frac{e_1 : \mathbf{nz} \quad e_2 : \mathbf{nz}}{e_1 * e_2 : \mathbf{nz}} \text{TYPEMULNZ} \qquad \frac{e_1 : \mathbf{mz} \quad e_2 : \mathbf{mz}}{e_1 * e_2 : \mathbf{mz}} \text{TYPEMULMZ} \\
\\
\frac{e_1 : \mathbf{mz} \quad e_2 : \mathbf{nz}}{e_1 / e_2 : \mathbf{mz}} \text{Typediv}
\end{array}$$

Justification: the sums and differences of nonzero integers can be zero (e.g. $1 + (-1)$, $1 - 1$), so the addition and subtraction can only give **mz**. A product of two nonzero integers is nonzero, but a product with a possibly-zero factor may be zero. For division, the divisor must be **nz** to avoid dividing by zero; the result is **mz** because integer division can produce zero even with a nonzero dividend (e.g. $1/2 = 0$).

Task 8

Progress. If $e : \tau$, then e val or there exists e' such that $e \mapsto e'$.

Proof. By induction on the derivation of $e : \tau$.

- **Case TypeNum.** Then $e = n$, and n val by VALNUM.
- **Case TypeNumNZ.** Then $e = n$ (with $n \neq 0$), and n val by VALNUM.
- **Case TypeDiv.** Then $e = e_1 / e_2$ with $e_1 : \mathbf{mz}$ and $e_2 : \mathbf{nz}$. By the induction hypothesis on $e_1 : \mathbf{mz}$, either $e_1 \mapsto e'_1$ or e_1 val.
 - If $e_1 \mapsto e'_1$, then $e_1 / e_2 \mapsto e'_1 / e_2$ by STEPSEARCHOPLEFT (with $op = /$).
 - If e_1 val, then by Canonical Forms (case 2), $e_1 = n_1$ for some n_1 . By the induction hypothesis on $e_2 : \mathbf{nz}$, either $e_2 \mapsto e'_2$ or e_2 val.
 - * If $e_2 \mapsto e'_2$, then $n_1 / e_2 \mapsto n_1 / e'_2$ by STEPSEARCHOPRIGHT (with $op = /$).
 - * If e_2 val, then since $e_2 : \mathbf{nz}$, Canonical Forms (case 1) gives $e_2 = n_2$ for some $n_2 \neq 0$. Hence the premise of STEPDIV holds, and $n_1 / n_2 \mapsto n$ where n is the integer quotient of n_1 by n_2 .

In every subcase e steps. □

Preservation. If $e : \tau$ and $e \mapsto e'$, then $e' : \tau$.

Proof. By induction on the derivation of $e \mapsto e'$.

- **Case StepAdd.** Then $e = n_1 + n_2$ and $e' = n$ where $n = n_1 + n_2$. The only typing rule whose conclusion has the form $e_1 + e_2$ is TYPEADD, so by inversion $\tau = \mathbf{mz}$. By TYPENUM, $n : \mathbf{mz}$, i.e. $e' : \tau$.

- **Case StepSub.** Then $e = n_1 - n_2$ and $e' = n$ where $n = n_1 - n_2$. The only typing rule for $e_1 - e_2$ is `TYPE SUB`, so by inversion $\tau = \text{mz}$. By `TYPE NUM`, $n : \text{mz}$.
- **Case StepDiv.** Then $e = n_1/n_2$ with $n_2 \neq 0$ and $e' = n$, the integer quotient of n_1 by n_2 . The only typing rule for e_1/e_2 is `TYPE DIV`, so by inversion $\tau = \text{mz}$. By `TYPE NUM`, $n : \text{mz}$. $\square$

Bonus Task 1

Case StepMul. Then $e = n_1 * n_2$ and $e' = n$ where $n = n_1 n_2$. Two typing rules have a conclusion of the form $e_1 * e_2$, so by inversion there are two cases.

- **Subcase TypeMulMZ.** Then $\tau = \text{mz}$, and $n : \text{mz}$ by `TYPE NUM`.
- **Subcase TypeMulNZ.** Then $\tau = \text{nz}$, with $n_1 : \text{nz}$ and $n_2 : \text{nz}$. Since $n_1 \text{ val}$ and $n_2 \text{ val}$ (by `VAL NUM`), Canonical Forms (case 1) gives $n_1 \neq 0$ and $n_2 \neq 0$. The product of two nonzero integers is nonzero, so $n = n_1 n_2 \neq 0$, and by `TYPE NUM NZ`, $n : \text{nz}$. $\square$

Task 9

Expression. $1/(1+1)$.

This never divides by zero (it evaluates to $1/2 \mapsto 0$), but it is not well-typed: the only rule for $/$ is `TYPE DIV`, which needs $1+1 : \text{nz}$, and the only rule for $+$ (`TYPE ADSD`) gives type `mz`.

Suggested refinement. Track signs. Replace `nz` with two more precise types `pos` (strictly positive) and `neg` (strictly negative), both of which count as nonzero (e.g. `pos` $\leq$ `nz` and `neg` $\leq$ `nz` in a subtyping sense). Then add rules such as

$$\frac{n > 0}{n : \text{pos}} \quad \frac{e_1 : \text{pos} \quad e_2 : \text{pos}}{e_1 + e_2 : \text{pos}} \quad \frac{e_1 : \text{neg} \quad e_2 : \text{neg}}{e_1 + e_2 : \text{neg}} \quad \frac{e_1 : \text{pos} \quad e_2 : \text{neg}}{e_1 - e_2 : \text{pos}}$$

together with the usual sign rules for multiplication (`pos * neg : neg`, etc.). Now $1+1 : \text{pos}$, hence $1+1 : \text{nz}$, and $1/(1+1) : \text{mz}$ by `TYPE DIV`. Division by zero is still ruled out, since every `pos` or `neg` expression evaluates to a nonzero integer. (This still rejects some safe programs, such as $1/(2-1)$; a more precise system could track exact values or intervals, e.g. refinement types like $\{x : \text{int} \mid x \geq 1\}$.)

Wrap-up

Task 10

Approximately 6 hours.